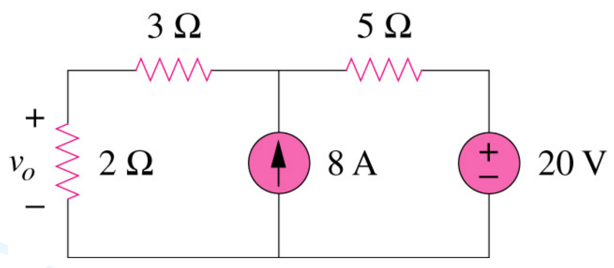
Circuit Theorems - Chapter 4

4.1 Motivation

If you are given the following circuit, are there any other alternative(s) to determine the voltage across 2Ω resistor?



What are they? And how?

Can you work it out by inspection?

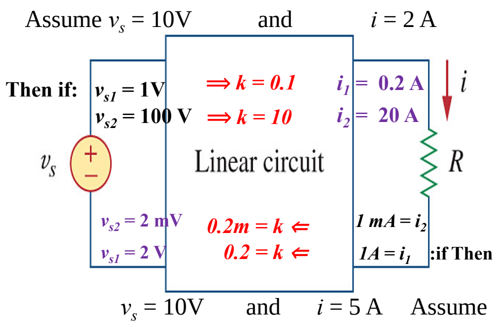
4.2 Linearity Property

It is the property of an element describing a linear relationship between cause and effect.

A linear circuit is one whose **output** is linearly related (or directly proportional) to its **input**.

Two property of linearity

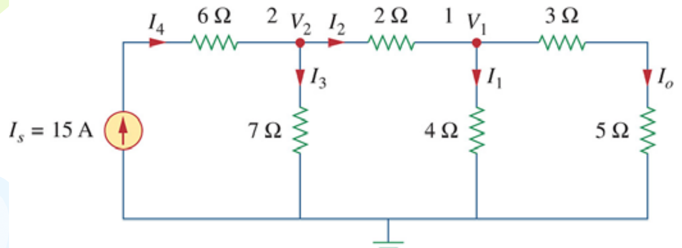
1. **Homogeneity (scaling):**
   $v = Ri$
   $\rightarrow kv = kRi$



2. **Additive (Superposition):**
   $v_1 = Ri_1$ and $v_2 = Ri_2$
   $v = v_1 + v_2 = (i_1 + i_2)R$

Example 4.2

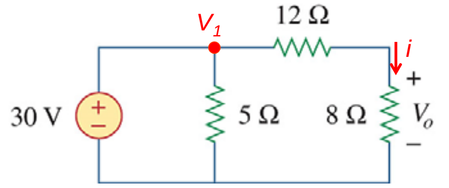
Assume I_o = 1 A. Use linearity to find the relationship between I_o and I_S. If I_S = 15A, what is the actual value of I_o in the circuit?



1. If I_o = 1 A: V_1 = 1×(3+5) = 8 V.
2. With V_1 = 8 V: I_1 = 8/4 = 2 A.
3. With I_1 = 2 A: I_2 = I_o + I_1 = 1+2 = 3 A (KCL)
4. With I_2 = 3 A: V_2 = V_1 + I_2×2 = 8+6 = 14 V $\quad I_2 = \frac{V_2 - V_1}{2}$
5. With V_2 = 14 V: I_3 = 14/7 = 2 A
6. With I_3 = 2 A: I_4 = I_2+ I_3 = 3 + 2 = 5 A (KCL)
7. Thus: $I_o/I_4 = I_o/I_S$ = 1/5 = k
8. Due to linearity: $I_{o\_actual} = k * I_{S\_actual}$ = 3 A

PP 4.2

Use linearity to calculate the relationship between V_1 and V_o. If V_1=30 V, what is the actual value of V_0?



Using voltage division

$$V_o = V_1 \frac{8}{12+8} = 0.4\,V_1 \Longrightarrow \boldsymbol{k = 0.4}$$

If V_1 equals 30 V, then according to linearity principle:

$$V_o = 0.4\,V_1 \Longrightarrow\ = 0.4 \times 30 = 12\ V$$

4.3 Superposition Theorem

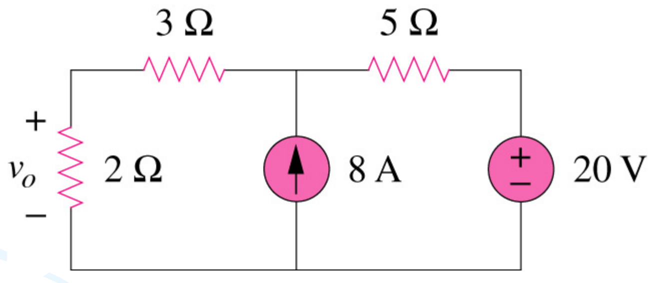
The voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltage across (or currents through) that element due to EACH independent source acting alone.

- $v = v_1 + v_2 + \ldots + v_n$
- $i = i_1 + i_2 + \ldots + i_n$

The principle of superposition helps us to analyze a linear circuit with more than one independent source by calculating the contribution of each independent source separately.

4.3 Superposition Theorem

We consider the effects of 8A and 20V one by one, then add the two effects together for final v_o.



$$v_o = v_{8A} + v_{20V}$$

4.3 Superposition Theorem

STEPS

Steps to apply superposition principle

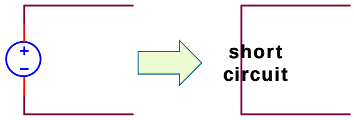
1. Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using nodal or mesh analysis.
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

4.3 Superposition Theorem

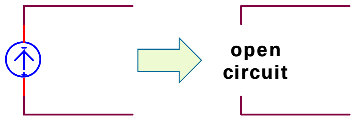
Zeroing of Sources

How do you turn off a source, i.e. make it zero?

1. Independent sources:
   - Independent voltage sources are replaced by 0 V (short circuit (R=0 Ω))

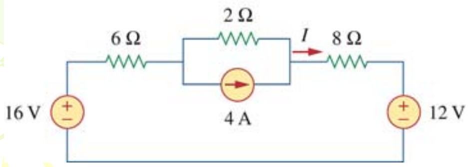


   - Independent current sources are replaced by 0 A (open circuit (R → ∞ Ω)).



2. Dependent sources are left intact because they are controlled by circuit variables.

PP 4.5: Use superposition to find *I*.



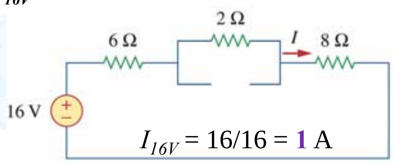
$I = I_{16V} + I_{12V} + I_{4A}$

I_{16V} – *current due to 16 V source*

I_{12V} – *current due to 12 V source*

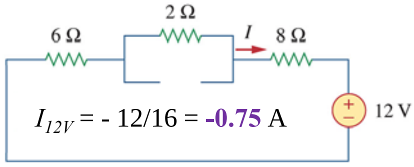
I_{4A} – *current due to 4 A source*

I_{16V} :Switch off the 4 A and 12 V sources



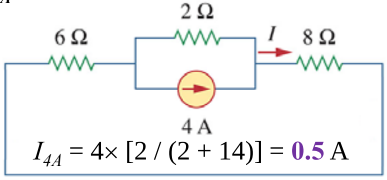
$I_{16V} = 16/16 = 1$ A

I_{12V} :Switch off the 4 A and 16 V sources



$I_{12V} = -12/16 = -0.75$ A

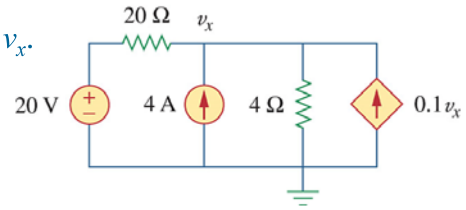
I_{4A} : Switch off the 12 V and 16 V sources



$I_{4A} = 4\times [2 / (2 + 14)] = 0.5$ A

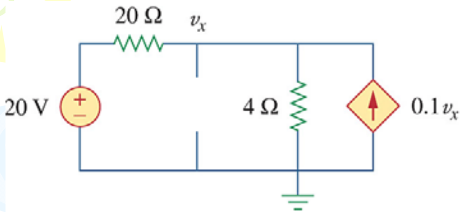
$$I = I_{16V} + I_{12V} + I_{4A}$$
$$= 1 + (-0.75) + 0.5$$
$$= 0.75 \text{ A}$$

PP 4.4: Use superposition to find v_x.



$\boxed{v_x = v_{x20V} + v_{x4A}}$

Switch off (zero) the 4 A source: v_{x20V}



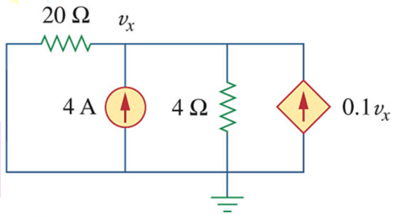
Nodal-analysis:

$$\frac{v_{x20V} - 20}{20} + \frac{v_{x20V}}{4} - 0.1v_{x20V} = 0$$

$$v_{x20V} - 20 + 5v_{x20V} - 2v_{x20V} = 0$$

$$v_{x20V} = 5\ V$$

Switch off (zero) the 20 V source: v_{x4A}



Nodal-analysis:

$$\frac{v_{x4A}}{20} - 4 + \frac{v_{x4A}}{4} - 0.1v_{x4A} = 0$$

$$v_{x4A} - 80 + 5v_{x4A} - 2v_{x4A} = 0$$

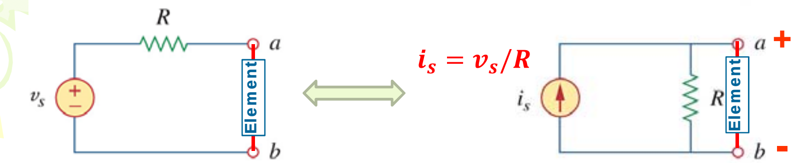
$$v_{x4A} = 20\ V$$

Total: $v_x = v_{x20V} + v_{x4A}$
$= 5 + 20 = \mathbf{25}$ V

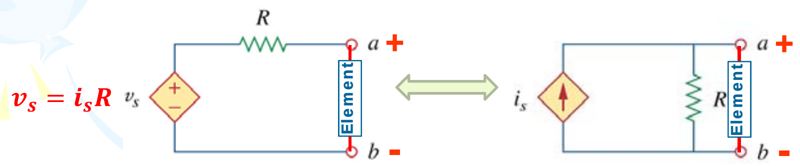
Ch. 4 L#2:
4.4 Source Transformations

- An equivalent circuit is one whose *v-i* characteristics are identical with the original circuit.
- It is the process of replacing a voltage source V_S in series with a resistor R by a current source I_S in parallel with a resistor R, or vice versa.

4.4 Source Transformation



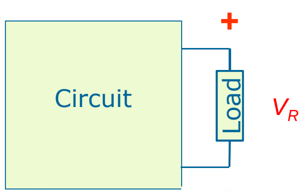
(a) Independent source transform

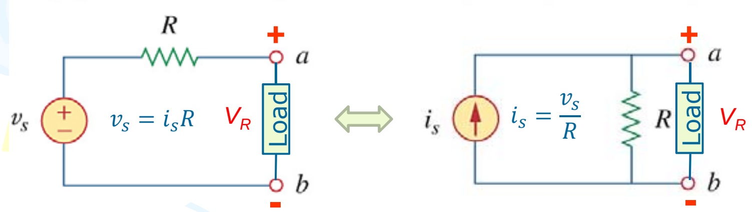


(b) Dependent source transform

4.4 Source Transformation

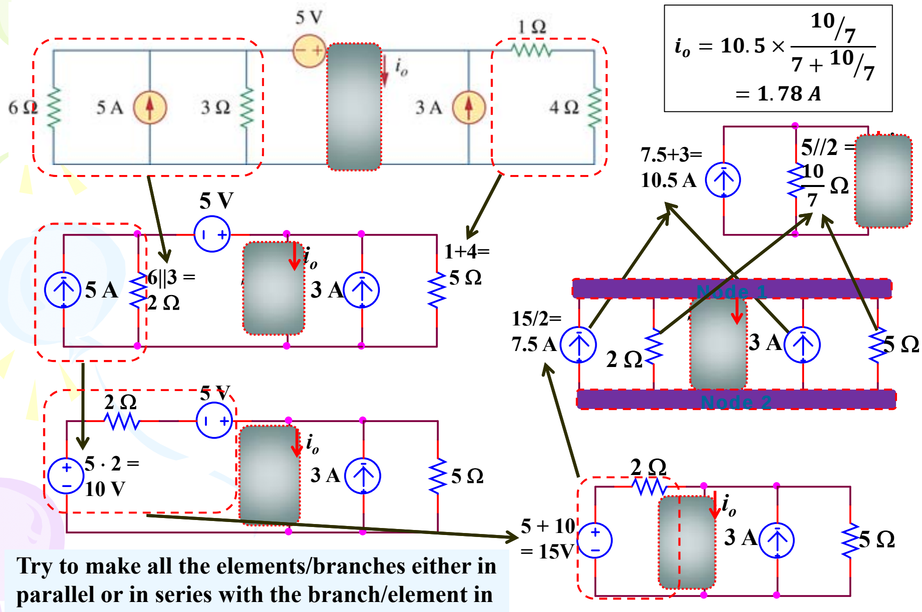
From a Load's perspective





A load connected between terminals *a-b* will not know the difference. The load will see the same voltage and conduct the same current. (I.e. V_R is the same for both circuits.)

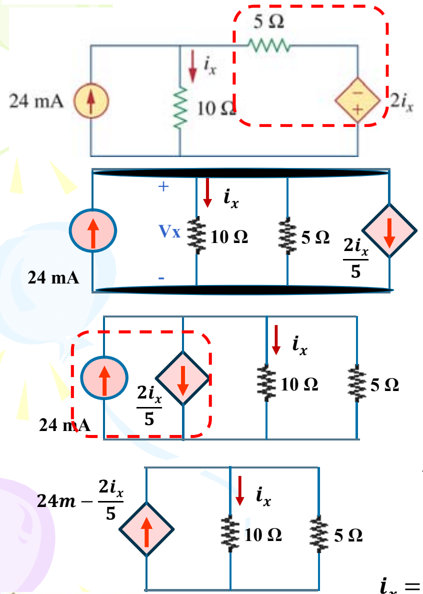
PP 4.5: Use source transformation and determine i_o.



$$i_o = 10.5 \times \frac{10/7}{7 + 10/7}$$
$$= 1.78\ A$$

Try to make all the elements/branches either in parallel or in series with the branch/element in question

PP 4.7: Use source transformation and determine i_x.



Very IMPORTANT:
Do NOT transform the part of the circuit with the unknown variable!

Method 2: Nodal Analysis

KCL: $-24m + i_x + \frac{v_x}{5} + \frac{2ix}{5} = 0$

$but\ vx = 10ix$

$\therefore (\times 5){:}\ 5ix + 10ix + 2ix = 120\,m$

$\therefore ix = 7.06\ mA$

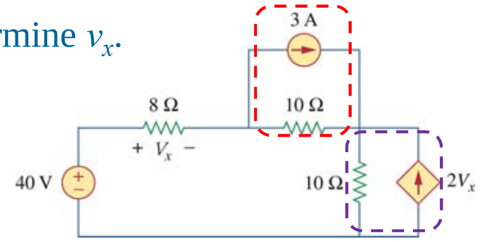
using current divider

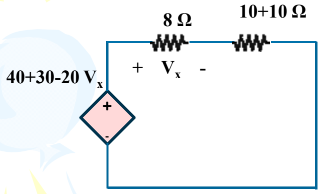
$$i_x = \frac{5}{10+5} \cdot \left(24m - \frac{2ix}{5}\right)$$

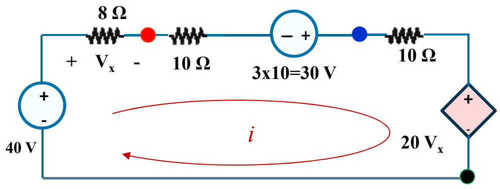
$$i_x = 7.06\ mA$$

Problem 4.24: Determine v_x.

Is it possible to make all the elements in single series loop?







Method 2: Voltage Divider

$$V_x = \frac{8}{8+20}(70 - 20Vx)$$

$$\therefore V_x = 2.98\ V$$

Method 1: KVL

$$-40 + V_x + 10i - 30 + 10i + 20\ Vx = 0$$

$$i = \frac{v_x}{8}$$

$$21V_x + 20\frac{v_x}{8} = 70$$

$$\therefore V_x = 2.98\ V$$

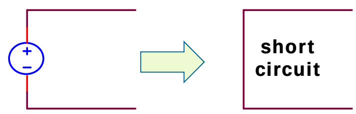
Summary

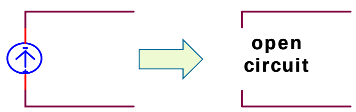
Superposition

- $v = v_1 + v_2 + \ldots + v_n$
- $i = i_1 + i_2 + \ldots + i_n$

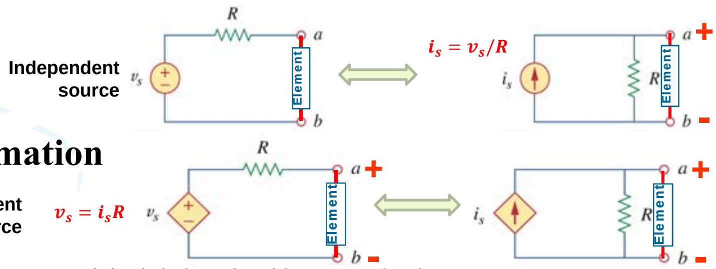
How do you turn off a source, i.e. make it zero?

1. Independent sources:
   Independent voltage sources are replaced by 0 V (short circuit (R=0 Ω))
   Independent current sources are replaced by 0 A (open circuit (R → ∞ Ω)).
2. Dependent sources are left intact because they are controlled by circuit variables.





Source Transformation

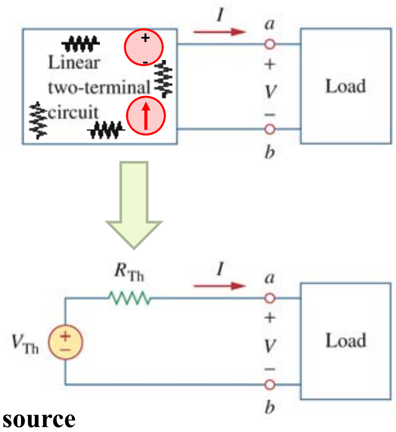


Ch. 4 L#3:
4.5 Thevenin's Theorem

A linear two terminal circuit can be replaced by an equivalent circuit consisting of voltage source (*V_{TH}*) in series with a resistor (*R_{TH}*), where

- *V_{TH}* is the open-circuit voltage at the terminals
- *R_{TH}* is the equivalent resistance at the terminal when all independent sources are off

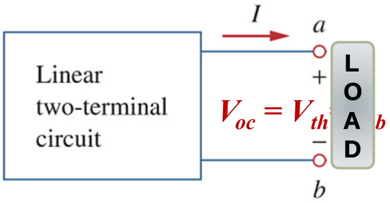
Terminals a-b are terminals between which you want to determine the Thevenins equivalent can be anywhere in the circuit.



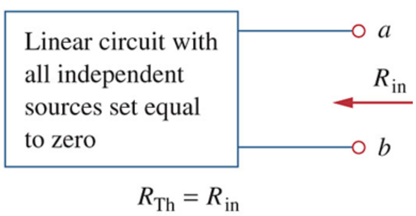
Case1: **Circuit with NO dependant source**
Case2: **Circuit with dependent source**

Case1: Circuit with NO dependant source
How to determine V_{th} and R_{th}

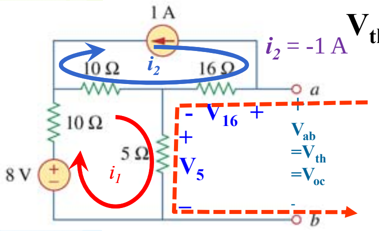
1. **Calculating V_{TH}:**
   - **Open-circuit, remove the element (load)**
   - **Calculate the voltage drop (potential difference) across the terminal (node) of the element (load).**



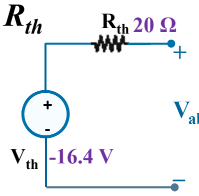
2. **Calculating R_{TH}:**
   - **Switch off all independent sources**
   - **Short-circuit the voltage source**
   - **Open-circuit the current source**
   - **Circuit should only have resistor**
   - **Calculate the $R_{eq}=R_{TH}=R_{in}$**



Problem 4-39: Determine the Thevenin equivalent circuit.



V_{th} & R_{th}



Note: Circuit is already short circuited. Or load is already removed

1. Calculating V_{TH}:
   - Open-circuit, remove the element (load)
   - Calculate the voltage drop (potential difference) across the terminal (node "ab") of the element (load).

$V_{th} = V_{ab} = V_{16} + V_5$

It is easier to use Mesh over Node

$= 16\, i_2 + 5\, i_1$

$= 16\,(-1) + 5\, i_1$

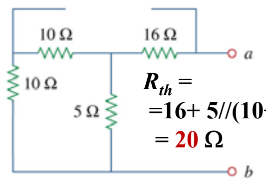
Loop 1: $-8 + 10i_1 + 10(i_1 - i_2) + 5i_1 = 0$

$25i_1 = -2 \qquad i_1 = -0.08$ A

$V_{th} = 16\,(-1) + 5\,(-0.08)$

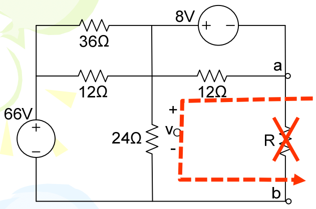
$\boxed{V_{th} = -16.4 \text{ V}}$

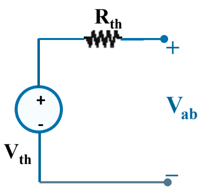
2. Calculating R_{TH}:
   - Switch off all independent sources
   - Short-circuit the voltage source
   - Open-circuit the current source
   - Circuit should only have resistor
   - Calculate the $R_{eq}=R_{TH}=R_{in}$



R_{th} = 16+ 5//(10+10) = 20 Ω

Calculate the Thévenin equivalent resistance R_{TH} across load R (terminal a-b).





1. Calculating V_{TH}:
   - Open-circuit, remove the element (load)
   - Calculate the voltage drop (potential difference) across the terminal (node "ab") of the element (load).

$V_{th} = V_{ab} = V_{12} + V_o$

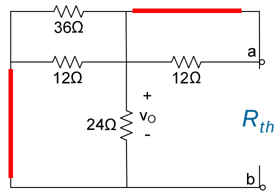
V_{12} = -8 V

Vo using voltage divider

$$V_o = 66\left(\frac{24}{24 + (36||12)}\right) = 48\text{V}$$

V_{th} = -8 + 48 = 40 V

2. Calculating R_{TH}:
   - Short-circuit the voltage source
   - Open-circuit the current source
   - Circuit should only have resistor



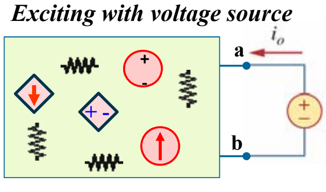
R_{th} = (12||36||24)
= 6.55
= 6.55 Ω

Case2: Circuit with dependent source

- Calculating V_{TH}:
  - *Same as that with Case1*
- Calculating $R_{TH}=R_{eq}$:
  - ➢ Switch off all independent sources
    - ➢ Short-circuit the voltage source;
    - ➢ Open-circuit the current source
  - ➢ Circuit should only have resistors and dependant source.
  - ➢ With dependant source present, it is trivial to calculate R_{eq}
    - ➢ We excite the open-terminal with a voltage source (v_o) or current source (i_o)
  - ➢ Since the circuit is linear

$$R_{TH} = \frac{v_o}{i_o}$$

Exciting with voltage source

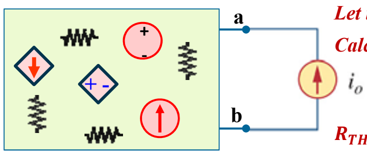


Let v_o=1V

Calculate i_o

$$R_{TH} = \frac{v_o}{i_o} = \frac{1}{i_o}$$

Exciting with current source



Let i_o=1 A

Calculate v_o

$$R_{TH} = \frac{v_o}{i_o} = \frac{v_o}{1}$$

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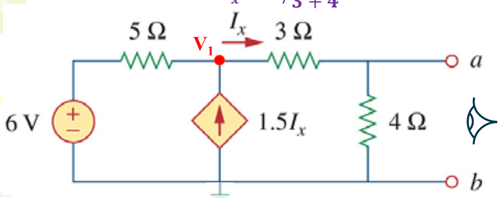
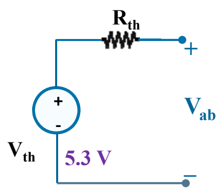
PP 4-9: Determine the Thevenin equivalent circuit to the left of terminals a-b.

$$I_x = \frac{V_1}{3+4}$$





Calculate $V_{th}=V_{oc}=V_{ab}=V_{4\Omega}$

KCL@V1

$$i_{5\Omega} - 1.5\,I_x + I_x = 0$$

$$i_{5\Omega} - 0.5\,I_x = 0$$

$$\frac{V_1 - 6}{5} - 0.5\frac{V_1}{3+4} = 0$$

Solve for V1

$$V_1 = 9.3\,V$$

Using voltage divider

$$V_{th} = \frac{4}{7}V_1$$

$$= 9.3 \times \frac{4}{7} = 5.3V$$

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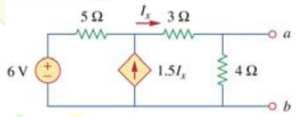
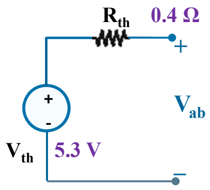
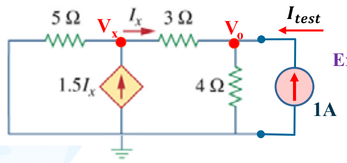
PP 4-9: continued $R_{th:}$ using CURRENT Test Source



Zero Ind. Sources





Excite using 1A current source

$$\therefore R_{th} = \frac{V_o}{I_{test}} = \frac{V_o}{1} = V_o$$

$= 0.4\,\Omega$

Calculate $V_{ab} = V_o$

KCL @ Vx

$i_5 - 1.5\, I_x + I_3 = 0$

$$\frac{V_x}{5} - 0.5 I_x = 0$$

$$\frac{V_x}{5} - 0.5\frac{V_x - V_o}{3} = 0$$

Simplify

$0.5V_x + 2.5V_o = 0$ (2)

KCL @ V_o

$i_{3\Omega} + i_{4\Omega} - 1 = 0$

$$\frac{V_o - V_x}{3} + \frac{V_o}{4} - 1 = 0$$

Simplify

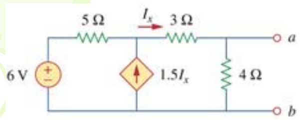
$-4V_x + 7V_o = 12$ (1)

(1) + 8 x (2) :

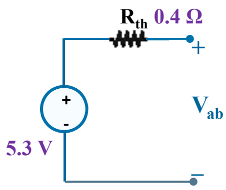
$27V_o = 12$

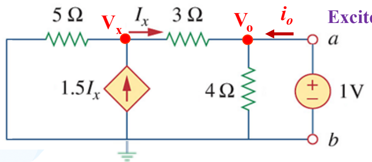
$V_o = 0.4\, V$

PP 4-9: continued $R_{th:}$ using - VOLTAGE Test Source $R_{th} = 0.4\ \Omega$



Zero Ind. Sources





Excite using 1V voltage source

$$R_{th} = \frac{V_{test}}{i_o} = \frac{1}{i_o}$$

$$= \frac{1}{2.25} = 0.4\,\Omega$$

Calculate i_o

KCL @Vx

$$\frac{V_x}{5} - 1.5I_x + \frac{V_x - 1}{3} = 0$$

But: $I_x = \dfrac{V_x - 1}{3}$

$$\frac{V_x}{5} - 1.5\frac{V_x - 1}{3} + \frac{V_x - 1}{3} = 0$$

Solve for Vx

$\therefore V_x = -5\, V$

KCL @Vo

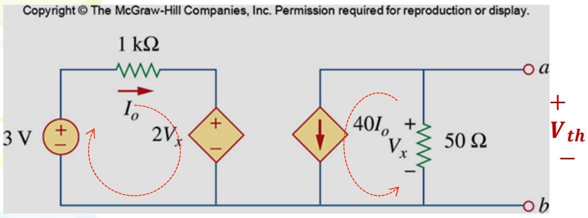
$$\frac{1 - V_x}{3} - i_o + \frac{1}{4} = 0$$

$4V_x + 12i_o = 7$

$4(-5) + 12i_o = 7$

$i_o = 2.25\, A$

P4.54: Find the Thevenin equivalent between terminal a-b



$$1000I_o + 2V_x - 3 = 0$$

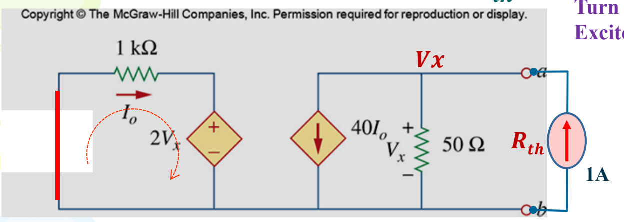
$$V_x = V_{th} = -40Io \times 50$$
$$= -2000Io$$

Solve for V_x

$V_x = 2V$

P4.54: R_{th}

Turn OFF all independent sources
Excite using 1A current source



$$1000I_o + 2V_x = 0 \quad (2)$$

$$R_{th} = \left(\frac{Vx}{1}\right) = Vx$$

KCL @ a

$$40Io + \left(\frac{V_x}{50}\right) = 1$$

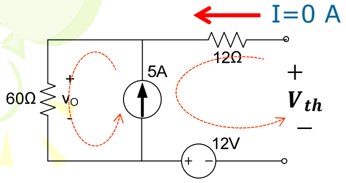
$$2000Io + Vx = 50 \quad (1)$$

Solve for Vx

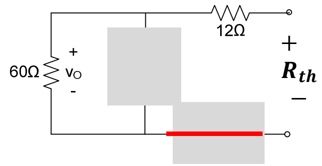
$V_x = -50/3V$

$R_{th} = -50/3\ \Omega$

Question: Determine the Thévenin equivalent.



Turn OFF all independent sources



$V_{th} = V_{12\Omega} + V_{5A} + 12$

$\Rightarrow V_{12\Omega}=0$

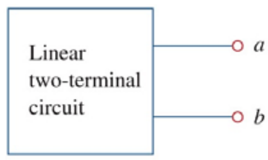
$V_{5A} = 60 \times 5 = 300$ V

$V_{th} = 300 + 12 = 312\ V$

$R_{th} = 12 + 60$
$= 72\ \Omega$

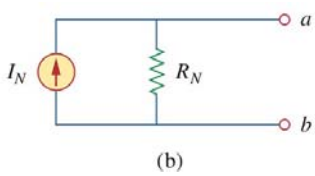
Ch. 4 L#4:
4.6 Norton's Theorem

It states that a linear two-terminal circuit can be replaced by an equivalent circuit of a current source I_N in parallel with a resistor R_N,



Where:

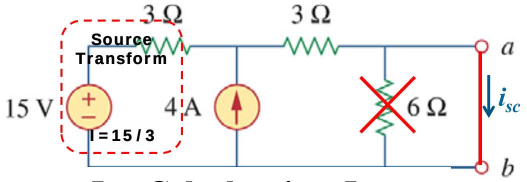
- Terminals a-b are terminals between which you want to determine the Norton equivalent.
- Can be anywhere in the circuit.
- I_N is the short circuit current (i_{sc}) through the terminals.
- R_N is the input or equivalent resistance at the terminals when the independent sources are turned off. (**R_N=R_{th}**)

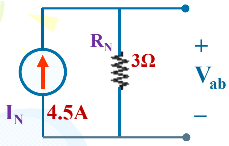


(b)

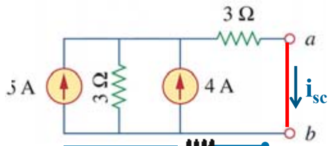
PP 4-11: Determine the Norton equivalent circuit to the left of terminals a-b.

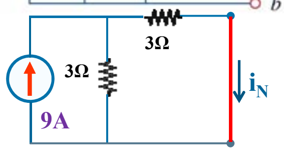
Case1: **Circuit with NO dependant source**



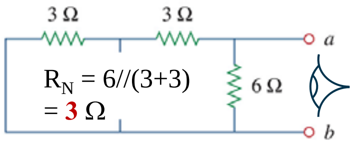


I_N: Calculate $i_{sc} = I_N$:





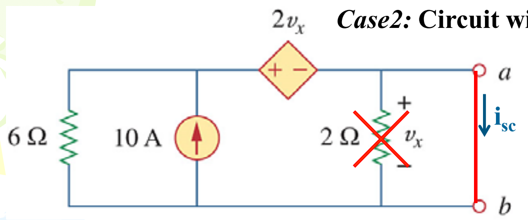
R_N: Zero ind. sources

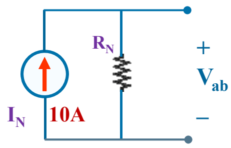


$I_N = 9/2 = 4.5$ A

PP 4-12: Determine the Norton equivalent circuit.

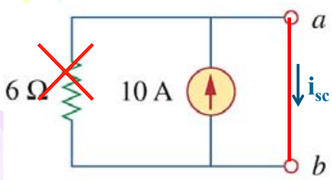
Case2: **Circuit with dependent source**





I_N: Calculate $i_{sc} = I_N$

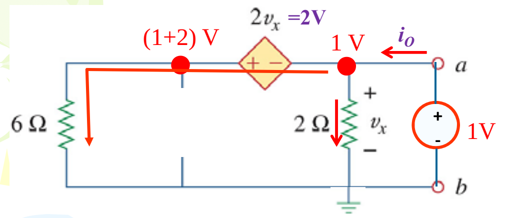
Observe: SC makes $v_x = 0$ V and hence also $2v_x = 0$.

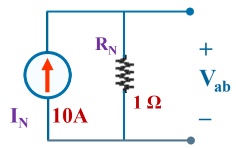


6 Ω shorted, hence no current through the 6 Ω:

$i_{sc} = 10$ **A**

PP 4-12: Determine the Norton equivalent circuit. R_N





R_N: Using Nodal analysis

Zero Ind. Sources + **1V** Test Source

Calculate i_o

KCL @ 1V node:

$$i_o = i_{2\Omega} + i_{6\Omega}$$

$$i_o = \frac{1}{2} + \frac{1+2}{6} = \mathbf{1}\ A$$

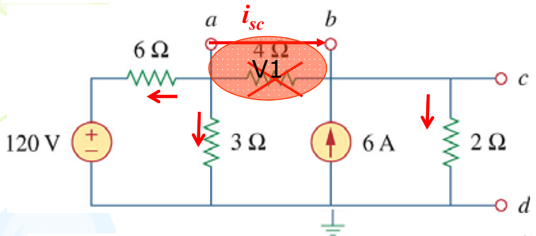
$$R_N = \frac{V_{test}}{i_o} = \frac{1}{i_o}$$

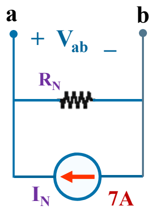
$$\boxed{= \frac{1}{1} = \mathbf{1}\ \Omega}$$

R_N: Repeat using 1 A source

Problem 4-51:

Determine the Norton equivalent circuit between terminals a-b.





Calculate i_{sc} = I_N

Method #1: Nodal Analysis

KCL @ V1:

$$i_6 + i_3 - 6 + i_2 = 0$$

$$\frac{V_1 - 120}{6} + \frac{V_1}{3} - 6 + \frac{V_1}{2} = 0$$

$$V_1 - 120 + 2V_1 - 36 + 3V_1 = 0$$

$$V_1 = \mathbf{26}\ V$$

KCL at node *b*:

$$-i_{sc} - 6 + i_2 = 0$$

$$-i_{sc} - 6 + \frac{V_1}{2} = 0$$

$$i_{sc} = I_N = -6 + \frac{26}{2} = \mathbf{7}\ A$$

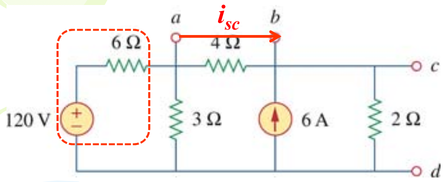
or KCL at node a:

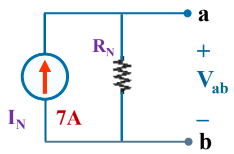
$$i_{sc} + i_6 + i_3 = 0$$

$$i_{sc} = -\frac{26-120}{6} - \frac{26}{3}$$

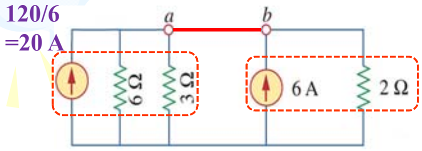
$$= \mathbf{7}\ A$$

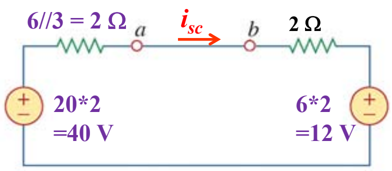
Problem 4-51: Method 2: I_N





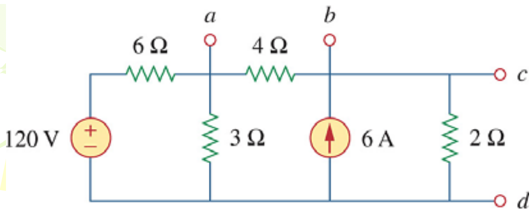
Calculate $i_{sc} = I_N$
Method #2: using Source Transformations

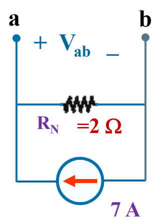




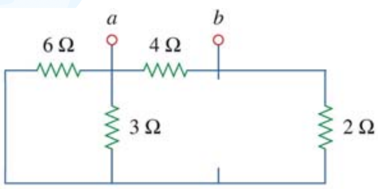
All in series:
i_{sc} = (40 – 12)/(2+2) = **7** A

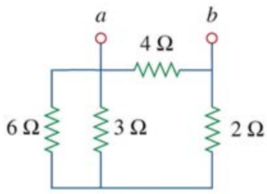
Problem 4-51: R_N





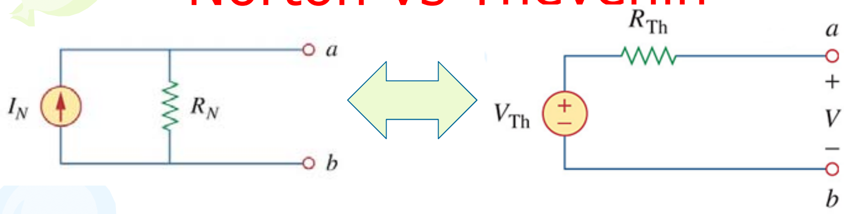
R_N: Zero Ind. Sources





R_{ab} = (6//3+2)||4
= (2+2)||4
= 4//4
= **2** Ω

4.6 Norton's Theorem
Norton vs Thevenin



Norton equivalent = Source transformation of the Thevenin equivalent and vice versa.

Where:

$$R_N = R_{th}$$
$$I_N = V_{th}/R_{th} = V_{th}/R_N$$
$$V_{th} = I_N R_{th} = I_N R_N$$

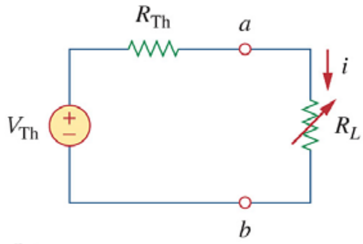
Ch. 4 L#5:
4.8 Maximum Power Transfer

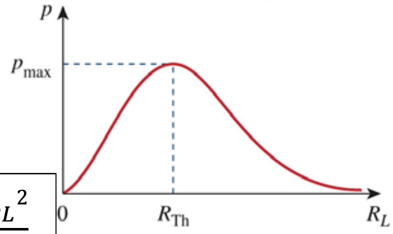
If the entire circuit is replaced by its <u>Thevenin equivalent</u> except for the load, the power delivered to the load is:

$$P = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L}\right)^2 R_L$$

For maximum power dissipated in R_L, P_{max}, for a given R_{TH} and V_{TH}:

$$\boxed{R_L = R_{th} \quad \rightarrow \quad P_{max} = \frac{{V_{th}}^2}{4R_{th}} = \frac{{V_{RL}}^2}{4R_L}}$$

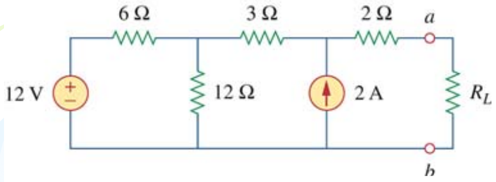


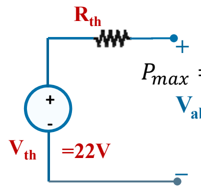


The power transfer profile with different R_L

Ex 4-13: Determine R_L for maximum power transfer as well as what P_{max} will be.

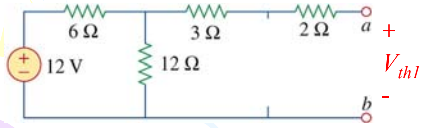
Method: Determine the Thevenin equivalents: V_{th} and R_{th}. between nodes "a-b"





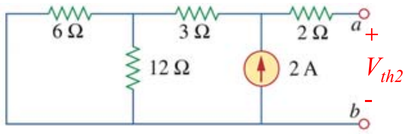
$$P_{max} = \frac{V_{th}^{\ 2}}{4R_{th}} = \frac{V_{RL}^{\ 2}}{4R_L}$$

V_{th}: using Super position:



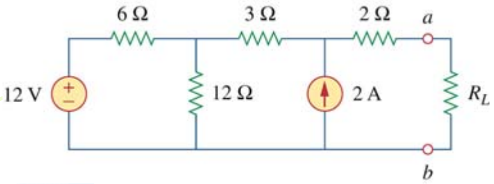
$V_{th1} = V_{12\Omega}$
$=12\text{ V} \times 12/(6+12) = 8\text{ V}$

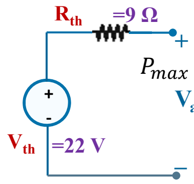
NO voltage drop across 3Ω and 2Ω resistor.



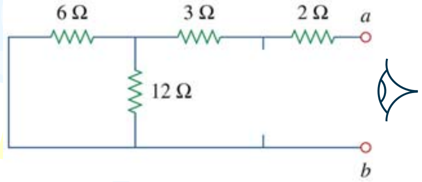
$V_{th2} = I \times R_{eq} = 2\text{A} \times ((6//12) + 3)$
$= 2 \times (4+3) = 14\text{ V}$
$\therefore V_{th} = V_{th1} + V_{th2} = 8 + 14$
$= \mathbf{22V}$

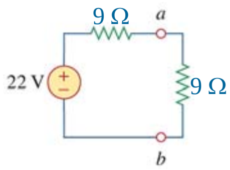
Ex 4-13: R_{th}: Zeroing of sources:





$$P_{max} = \frac{V_{th}^{\ 2}}{4R_{th}} = \frac{22^2}{4R_L}$$

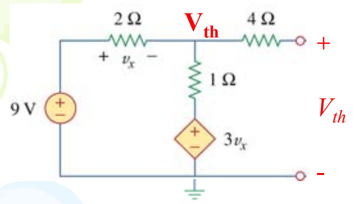


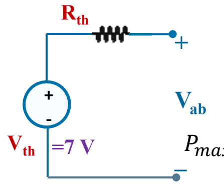


$R_{th} = 6//12 + 3 + 2$
$= \mathbf{9\ \Omega} = R_L$

$$\boxed{P_{max} = \frac{22^2}{4 \times 9} = \mathbf{13.44\ W}}$$

PP 4-13: Determine R_L for maximum power transfer as well as P_{max}





$$P_{max} = \frac{V_{th}{}^2}{4R_{th}}$$

$v_x = 9 - V_{th}$

KCL@V_{th}

$$\frac{V_{th} - 9}{2} + \frac{V_{th} - 3v_x}{1} = 0$$

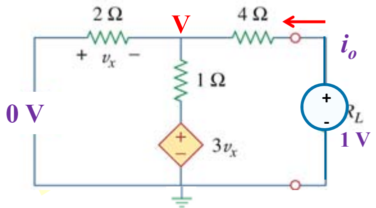
$$\frac{V_{th} - 9}{2} + \frac{V_{th} - 3(9 - V_{th})}{1} = 0$$

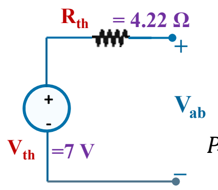
$$V_{th} - 9 + 2(V_{th} - 27 + 3Vth) = 0$$

$$V_{th} + 2Vth + 6Vth = 9 + 54$$

$$V_{th} = {}^{63}/_{9} = 7\ V$$

PP 4-13: R_{th} & P_{max}





$$P_{max} = \frac{7^2}{4R_{th}}$$

R_{th}: Zero 9V & Excite using 1V test source

Nodal: KCL@ V

$i_2+i_1+i_4=0$

$v_x = -V$

$$\text{x4}\{\ \frac{V}{2} + \frac{V - 3v_x}{1} + \frac{V - 1}{4} = 0$$

$$2V + 4(V - 3(-V)) + (V - 1) = 0$$

$$2V + 16V + V = 1$$

$$V = {}^{1}/_{19}$$

$$i_o = \frac{1 - V}{4} = 0.237\ A$$

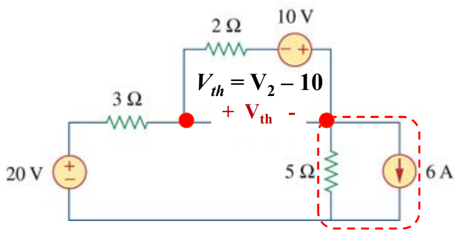
$$R_{th} = R_L = {}^{v_{test}}/_{i_o} = \frac{1}{0.237} = 4.22\ \Omega$$

$$\boxed{P_{max} = \frac{7^2}{4 \times 4.22} = 2.90\ W}$$

Problem 4.66:

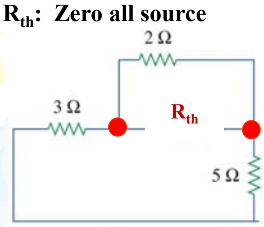
Determine the maximum power that can be delivered to resistor R.

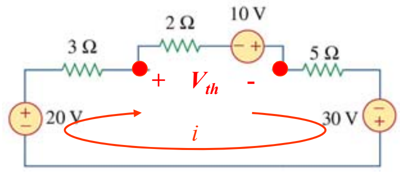
Max power is delivered to R when $R=R_{th}$

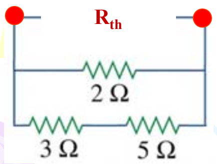


R_{th}: Zero all source

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$







R_{th} = 2 // (3+5) = 1.6 Ω

voltage divider

$$V_2 = \frac{2}{2+5+3}(20+10+30) = 12\ \text{V}$$

$$V_{th} = 12 - 10 = 2\ \text{V}$$

or

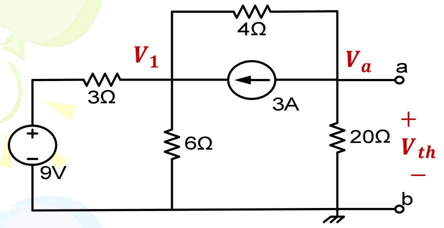
i = 20+10+30)/(3+2+5)= 6 A

V_{th} = 2 x 6 – 10 = 2 V

$$\boxed{P_{max} = \frac{2^2}{4 \times 1.6} = 0.625\ W}$$

Additional problems

Question: Determine V_{th} and R_{th} across terminal a-b?



$V_{th} = V_{20}$

KCL @ V_a

$i_{20} + 3 + i_4 = 0$

$$\frac{V_a}{20} + 3 + \frac{V_a - V_1}{4} = 0$$

$6Va - 5V_1 = -60 \quad (1)$

KCL @ V_1

$i_3 + i_3 + i_4 = 3$

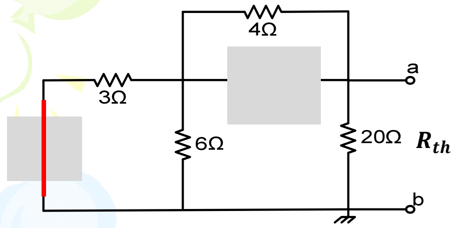
$$\frac{V_1 - 9}{3} + \frac{V_1}{6} + \frac{V_1 - Va}{4} = 0$$

$9V_1 - 3Va = 72 \quad (2)$

Solve (1) & (2)

$V_{th} = -4.62\ V$

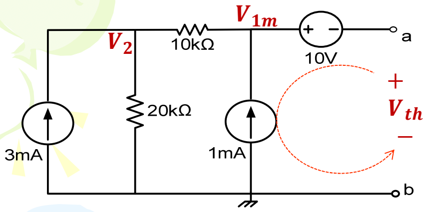
Question: R_{th}



Turn OFF all independent sources

$$R_{th} = (3||6 + 4)\ ||\ 20$$
$$= (2 + 4)\ ||\ 20$$
$$= 6\ ||\ 20$$
$$= 4.62\Omega$$

Question: Calculate V_{th} and R_{th} across terminal a-b?



$$V_{th} = -10 + V_{1m}$$

KCL @ V_1

$$-1m + \frac{V_{1m} - V_2}{10k} = 0$$

$$V_{1m} - V_2 = 10 \quad (1)$$

KCL @ V_2

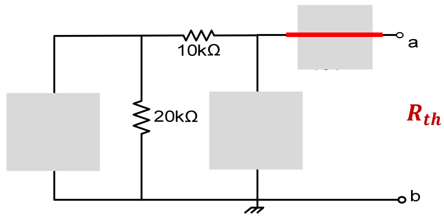
$$\frac{V_2 - V_{1m}}{10k} + \frac{V_2}{20k} - 3m = 0$$

$$-2V_{1m} + 3V_2 = 6 \quad (2)$$

Solve for V_{1m}

$$V_{1m} = 36\ V; \quad V_2 = 26\ V$$

$$V_{th} = -10 + 36 = 26\ V$$



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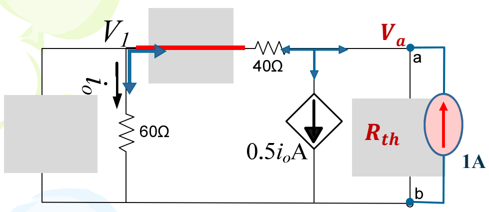
$$R_{th} = 10k + 20k$$
$$= 30\ k\Omega$$

Question: Find the Thévenin equivalent R_{TH} with respect to terminals a–b.



Remove the load between "ab"
Turn OFF all independent sources
Excite using 1A current source

$$R_{th} = \left(\frac{V_a}{1}\right) = Va \qquad R_{th} = 66.67\ \Omega$$

KCL @ a

$$0.5io + \left(\frac{V_a - V_1}{40}\right) = 1 \quad (1)$$

$$i_o = \frac{V_1}{60} \quad (2)$$

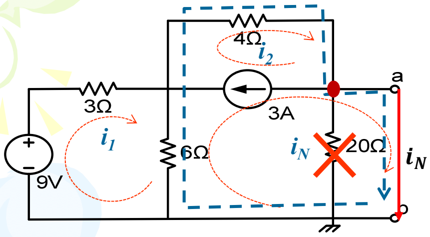
KCL @ V_1

$$\frac{V_1}{60} + \left(\frac{V_1 - Va}{40}\right) = 0 \quad (3)$$

Solve for V_a

$$V_a = 66.67\ V$$

Question: Determine the Norton current i_N for the circuit across terminal a-b?



$KLC@\ i1$

$-9 + 3i_1 + 6i_1 - 6iN = 0$

$9i_1 - 6iN = 9 \quad (1)$

$KLC@\ SM$

$4i_2 + 6iN - 6i_1 = 0 \quad (2)$

$KLC@\ CS$

$-i_2 + 3 + iN = 0$

$-i_2 + iN = -3 \quad (3)$

Solve for i_N

$i_N = -1\ A$